# RRAZ Depth

Installation and check · Affinity 3 · Windows 10/11 (64-bit) · version 1.0.0

This sheet covers installing RRAZ Depth and checking that it runs. Depth builds a 16-bit depth map from the active layer, on this computer's processor. The photographer's notes are a separate document, linked at the end. Everything runs on this computer: no upload, no account, no server.

> **These filters only work on 16-bit rasterised images.**
> A 16-bit per channel pixel layer, in RGB or greyscale. 8-bit and 32-bit HDR documents cannot be processed.

## 1 Install

### Before you start

- Windows 10 or 11, 64-bit.
- **No NVIDIA card and no CUDA required**: Depth runs on the processor, so any 64-bit Windows machine will do.
- About **1.5 GB free** while installing: the archive is about 321 MB and the expanded engine about 1 GB. You can delete the archive afterwards.
- Close Affinity. The plug-in cannot be replaced while Affinity is using it.
- The install is per user. No administrator rights are needed.

### The two files

Keep these two files together in one folder:

- `RRAZ_DEPTH_1.0.0_Setup.exe`
- `RRAZ_DEPTH_RUNTIME_1.0.0.7z`

Setup looks for the runtime archive in the **same folder** as the setup, for example both files in Downloads. The archive is not unzipped and is not inside the setup. If the archive is somewhere else, setup asks you to select it.

### Run the setup

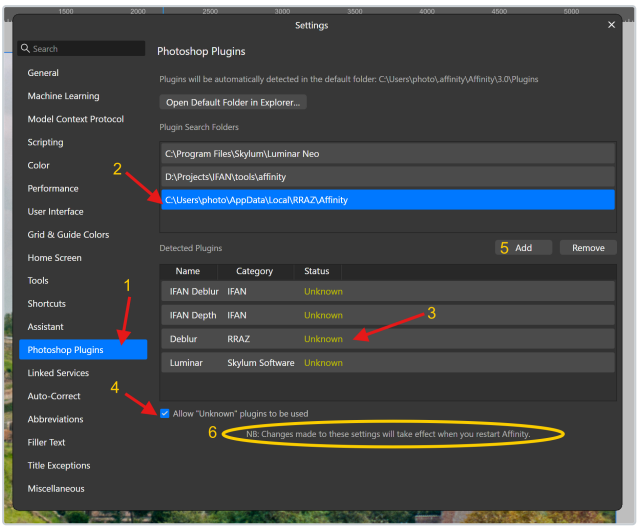
- Double-click the setup. Windows may warn that it is unsigned: choose **More info**, then **Run anyway**.
- Setup verifies the archive's SHA-256, then expands the engine. This takes a minute or two.
- No files are written outside `%LOCALAPPDATA%\RRAZ`.

### Check that it installed

- **Settings > Apps** lists **RRAZ Depth 1.0.0**.
- The engine is in `%LOCALAPPDATA%\RRAZ\Depth` (it contains `python\python.exe` and `app\`).
- The plug-in files are in `%LOCALAPPDATA%\RRAZ\Affinity`: `Depth.8bf`, `to_raw.py`, `depth.ini`.

### Add the folder to Affinity

Affinity's plug-in folder is a preference, so it has to be added once, the same way for every RRAZ filter. The setup finish page shows the exact folder for your Windows account.




*Affinity Settings > Photoshop Plugins (1). Other rows in this example are unrelated; only the RRAZ row matters.*

1. Open Affinity, then **Settings > Photoshop Plugins** (1).
2. Already have an RRAZ filter installed? If your RRAZ folder is listed in **Plugin Search Folders** (2), a filter is listed in **Detected Plugins** (3), and (4) is ticked, the folder is already added: close and reopen Affinity and go to step 6.
3. Otherwise tick **Allow "Unknown" plugins to be used** (4) and click **Add** (5).
4. Paste `%LOCALAPPDATA%\RRAZ\Affinity` into the folder dialog, press Enter, then choose **Select Folder**.
5. Close Affinity and open it again; the note at (6) says changes take effect on restart.
6. Open **Filters > Plugins > RRAZ > Depth**.

That folder is normally `C:\Users\<you>\AppData\Local\RRAZ\Affinity`, and the user name is different on each computer. Add the **Affinity** folder, not the engine folder (`...\RRAZ\Depth`). Any folder already listed can stay; RRAZ is a separate entry.

### To uninstall

**Settings > Apps > RRAZ Depth > Uninstall**. It removes its engine and files. The plug-in folder entry in Affinity is a preference and can be removed there.

## 2 Check it in Affinity

### Open a 16-bit image

- Use a 16-bit per channel, rasterised pixel layer: **RGB (16 bit)** or **Grey (16 bit)**.
- With an 8-bit or 32-bit HDR document the RRAZ Depth menu entry is **greyed out**. Convert first with **Document > Colour Format > RGB (16 bit)** (or Grey 16-bit).
- If the filter is run on an unsupported bit depth anyway, the layer becomes flat **mid-grey** — 50% grey (128 in 8-bit, 16384 in Affinity's 16-bit range, 0.5 in 32-bit). That is a refusal, not a result. Undo; the log says why.

### Run the filter

- Select the pixel layer and open **Filters > Plugins > RRAZ > Depth**.
- The active layer is replaced by the depth map. **Duplicate the layer first**: the original pixels are not kept.
- The map is 16-bit greyscale, returned at the layer's own size. The model works at a smaller internal size and scales the result up, so the map covers the whole frame with no crop.
- Affinity waits while it runs. On the processor a large layer takes noticeably longer than a small one. Everything is local; no image leaves the computer.

### What success looks like

- The same layer and dimensions, now a greyscale depth map. There is no success dialog.
- A line is added to `%LOCALAPPDATA%\RRAZ\Affinity\depth.log`.

### If the menu entry is missing

- Check that `%LOCALAPPDATA%\RRAZ\Affinity` is listed in **Settings > Photoshop Plugins**.
- Restart Affinity.
- Read `%LOCALAPPDATA%\RRAZ\Affinity\depth.log`.
- Confirm the layer is 16-bit, not 8-bit or 32-bit.

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**Depth model:** *Depth Anything At Any Condition* (Depth Anything AC), HVision-NKU, built on Depth Anything V2. Model code and weights are licensed CC BY-NC 4.0 with Apache-2.0 components; the licence files are installed in `app\LICENSES\`. Original code: github.com/HVision-NKU/DepthAnythingAC.

**Changes for the Affinity plug-in:** github.com/RRaz/Filters/. Not affiliated with or endorsed by the original authors.

**All RRAZ filters, notes and example images**
https://github.com/RRaz/Filters/

The list, notes and images are kept separately and updated independently of this installer. Download them when you want; there is no ZIP.